Engineering Mechanics 1 · Unit_01 · UNIT 1

# DRAFT — UE26ME142A · Unit_01 · UNIT 1 · UE26ME142A Unit 1 Class 9 AVSummary

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

# · How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

| | |
|---|---|
| Source text | Verbatim or OCR-derived text, kept separate from editorial notes. |
| Candidate | Automated topic/source association requiring manual confirmation. |
| Low ASR confidence | Transcript segment requires audio verification. |
| DRAFT | Human review remains incomplete. |

## Contents

1. 1 · UE26ME142A Unit 1 Class 9 AVSummary
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

# UE26ME142A Unit 1 Class 9 AVSummary

## 1 UE26ME142A Unit 1 Class 9 AVSummary

Study focus: UE26ME142A_Unit 1_Class 9_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000182 · EV0000183

SHORT-NOTE RECAP

**Core explanation · 00:00–04:59**

My name is Dr.

**Key details**

- SV Venkitish, Civil Engineering Department of Science and Humanities, PS University.
- This is class number 11.
- In the last class we discussed about couple, the effects of couple, the classification of couple, the characteristics of couple, force couple system, then we solved a few problems.
- Today, that is in class number eleven, we will try to solve few more unsolved problems from the prescribed textbook that is Engineer Mechanics Tactics by Miriam, Craig and Balden.
- Use in this problem, which is 2 bar 68, the statement of the problem is the force F is applied at the end of ACD, which is mounted to a vertical post.
- Replace this single force F by an equivalent force couple system at B.
- Next redistribute this force and couple by replacing it with two forces.
- acting in the same direction as F, one at C and the other at D, and determine the forces supported by the two X bolts.
- Using values of F equals 425 newtons, theta equals 30 degrees, B equals 1.9 meters, D equals 0.2 meters.
- H equals point eight meters and I equals two point seven five meters.
- There is a force that is acting at the end of the arm A C D, that is at this point A.
- We have to replace this force by an equivalent force couple system at B, that is at this point.
- Redistribute this force and couple by replacing it with two forces acting in the same direction as F.
- Same direction of F means it is at this angle theta.
- He has not specified the sense of the force.

- He has specified that the angle of the force that is going to be acting one at C and one at B should have the same direction, which means it is at angle T with respect to the vertical.
- and determine the forces supported by the two X bolts.
- X boles is these are the two bolts that is C and D.
- What are the forces acting at that?
- Then the angle theta is 30 degrees.
- Now, first thing is we have to replace this force by a force couple system at B.
- I will find out the component of this force and use that as a force that is going to be acting at B.
- P is equal to the component of this force here along X would be to the left.
- I am taking.
- X axis to the right as positive and upwards y axis as positive.
- Then component of this force along Y will be downwards, so again negative.
- and it's going to be a cause of this angle theta j.

**Example or application from the lecture**

- And then he has given you the values of the force F which is 425 newtons.
- Then the B distance that is from A to the center between the two volts that is C and D is given as 1.9 meters.
- Then the distance D that is from the center of the bolts C and D till the C, R T is given as D and D.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

**Core explanation · 05:00–09:59**

Minus three sixty eight point zero six J Kilo Newtons because sorry, if this is in Newtons, I'm sorry this has to be changed to newtons because what is given here is in newtons.

**Key details**

- So once we find that, the next thing that we have to do is we have to determine the moment about B because of this force that is acting.
- So moment about P because of the horizontal component will be F sine theta into L minus H because the horizontal component vertical distance is L minus H.
- Then we will take the vertical component which is F cos teta into the horizontal distance is V.
- Straight away we can put that, then substitute the value for F, L, H and B, we get the moment about B.
- which is equal to F sine theta i with a negative sign minus f cos theta j which is acting along the y axis that is the force then movement we have already got because of the two components and multiplying it with the respective distances perpendicular distances.
- The second part of the problem is to replace this force and the couple by two forces acting at C and D, which will have the same direction, that means this direction.
- The movement acting at P will be in the anti-clockwise direction.

- That should produce a movement at B in the anticlockwise direction.
- Then when I apply the force at D, that should also produce a movement at B in the anticlockwise direction.
- This force when I apply, this force should produce a moment at B which is in the anti-clockwise direction.
- That would be appropriate as what I feel.
- Or F B and also the movement at B, whatever is there, should be the movement that is going to be produced by these two forces.
- To be equal to 0.
- So we will get this force F C which is acting downwards minus F C which is acting upwards should be equal to F because this is the total force that should be.
- That we should get when we sum up these two, so that I have equated, and f we know that it is equal to 425 newtons.
- No momentum or B whatever we get because of these two.
- That would produce an anti-clockwise movement, so positive.
- Then the component of F C in the horizontal direction, that is F C sin theta into the vertical distance from B to this point B to C, which is L minus H, and that will also produce an anti-clock movement, so positive.
- Next, the component of the force along vertical, which is because of Fd, would be Fd.
- cos teta into D into D which will be positive because that will also produce an anti-clockwise movement.
- But the component of F T acting in the horizontal direction that is going to be this would be the sense that will present clockwise movement.
- Sine theta into the horizontal vertical distance is L minus H, so that is what is indicated here.

**Example or application from the lecture**

- Okay, that's the first part of the problem to replace the given force which is acting at A point to a force couple system acting at B.
- We are taking so that would be equal to F C, which is this component of this acting along the vertical, which is F C cos theta into D, that is what is given here.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000186 · EV0000187 · EV0000188 · +1 source units in pack.json

**Core explanation · 10:00–14:59**

Then theta is 30 degrees and D we all know it is 0.2.

**Key details**

- From this we would be able to get the value of F D which is equal eight hundred and seven newtons.
- I hope all of you have understood.
- You have to replace that by a force couple system at B.
- Then replace this force couple system by.

- Forces acting at C and D is what we have done.
- Next, another problem we will discuss, that is problem number 2 bar 76.
- Seal seed back release mechanism.
- The part is subjected to the four newton force exerted at A and a 300 newton millimeter restoring moment exerted by a hidden torsional spring.
- Determine the y-intercept of the line of action of the single equivalent force.
- And there is a couple or a moment of 300 newton millimeters that is acting.
- We have to replace this by a single equivalent force.
- And we should find out where this single equivalent force should act along the y-axis so that produces the same effect as that of the single force of and the moment 300 newton millimeters.
- This one, this problem is.
- Opposite of that of the force couple system.
- Then we can also note that the point A is at a distance of 10 millimeters from O horizontally.
- and it is at a distance of 40 millimeters from O vertically.
- What is that single equivalent force that should be applied on the y axis so that it produces the same effect.
- 40 newtons, but we have to apply this force on the y-axis such that force when I apply it on the y-axis would produce a movement which will be the combination of the movement produced by F acting at A and this movement 300 newtons.
- So, first thing that we need to find is what would be the combined moment by this force F and the moment 300 newtons that would be acting.
- So first thing is what would be the moment M naught.

**Example or application from the lecture**

- What we used to do earlier was there was one force given, we used to determine what should be the force couple system, which would produce the same effect when it is applied at some other point, is what we used to determine.
- Going forward, note down what is given, the force F is given, the moment is given.
- From the diagram, we can note down that the angle at which the force is acting with respect to the horizontal axis is 15 degrees, that is also given.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000190 · EV0000191 · EV0000192 · +1 source units in pack.json

**Core explanation · 15:00–19:59**

Combination of this and this is what we have to determine.

**Key details**

- Sine theta by theta is 15 degrees, so this component we will get into this distance 10.
- So we will get the moment because of this component along y.
- So movement because of the component along X would be in the clockwise direction, so it is negative.

- The component will be of cos theta into the vertical distance which is 40 mm.
- Only the component at the horizontal component would produce a moment.
- Cos 15 into y should be equal to the moment at M naught.
- I hope all of you have understood.
- The two problems that I have discussed today, which are not solved in the textbook, I have exclusively taken these two problems, which are.
- Tougher and not easy to understand.
- I hope all of you have got this problems.
- This is class number 11.
- We will be continuing the class.
- My name is Dr.
- I am from Civil Engineering under the Department of Science and Humanities, which comes under PS University.
- See you in the next class.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000194 · EV0000195 · EV0000196 · +1 source units in pack.json

# A Symbol glossary

| Symbol | Name | Meaning | Appears in |
|---|---|---|---|

# B Sources, coverage and gaps

| Source | Path | Use |
|---|---|---|
| UE26ME142A_Unit 1_Class 9_AVSummary transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/009_UE26ME142A_Unit_1_Class_9_AVSummary__866a19feaac2a3bf.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| UE26ME142A_Unit 1_Class 9_AVSummary transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/009_UE26ME142A_Unit_1_Class_9_AVSummary__866a19feaac2a3bf.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |

| Source | Path | Use |
|---|---|---|
| UE26ME142A_Unit 1_Class 9_AVSummary transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/009_UE26ME142A_Unit_1_Class_9_AVSummary__866a19feaac2a3bf.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps. |
| UE26ME142A_Unit 1_Class 9_AVSummary transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/009_UE26ME142A_Unit_1_Class_9_AVSummary__866a19feaac2a3bf.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

## Known gaps, conflicts and corrections

1. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md have no readable `.slides.md` text; their original page images are included.
4. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md have no readable `.slides.md` text; their original page images are included.
5. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md have no readable `.slides.md` text; their original page images are included.
7. 3 rendered pages in 004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md have no readable `.slides.md` text; their original page images are included.
8. 3 rendered pages in 005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md have no readable `.slides.md` text; their original page images are included.

9. 2 rendered pages in 006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md have no readable `.slides.md` text; their original page images are included.
10. 3 rendered pages in 008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md have no readable `.slides.md` text; their original page images are included.
12. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.md have no readable `.slides.md` text; their original page images are included.
14. 2 rendered pages in 011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md have no readable `.slides.md` text; their original page images are included.
16. 4 rendered pages in 012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md have no readable `.slides.md` text; their original page images are included.
17. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
18. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
19. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
20. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
21. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

## Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

| Source | Unit counts | Example decisions |
| --- | --- | --- |
| SRC00042 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare... |
| SRC00043 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare... |

| Source | Unit counts | Example decisions |
|---|---|---|
| SRC00044 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare... |
| SRC00045 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare... |

# C Revision and retrieval practice